

Maggie Matabane

Data Analytics (May intake)

Exercise 02: SQL Fundamentals

1. Syntax: `SELECT department`
`FROM students-table;`

Results:

Students-table
department
IT
HR
Finance

2. Syntax `SELECT AVG(age) AS Avg-age,`
`department`
`FROM students-table`
`GROUP BY department;`

Results

Students - table	
Avg-age	department
30.5	IT
22	HR
23	Finance

3. Syntax `SELECT department,`
`Count (Student-id) AS student-count`
`FROM student-table`
`GROUP BY department`
`Having student-count > 2;`

Results

5

Students-table	
department	student-count
IT	2
HR	2

4) Syntax SELECT student_id,
name,
age,
department

from student-table

WHERE age BETWEEN 21 AND 23;

Student-table

Student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5, Syntax SELECT student-id,
name,
age,
department

FROM student-table

WHERE department = 'IT' OR department = 'HR' AND age > 21;

Student-table

Student-Id	name	age	department
2	Bob	22	HR
5	Diana Eve	22	HR

6. Syntax

```
SELECT department,  
sum(credits) AS total-credits  
FROM courses-table  
GROUP BY departments  
HAVING sum(credits) > 5;
```

Results.

Courses-table

department	total-credits
IT	11

7. Syntax

```
SELECT course_id,
       course_name,
       department,
       credits
FROM   courses-table
WHERE  credits < 4;
```

Results: courses-table

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Syntax

```
SELECT course_id,
       course_name,
       credits
FROM   courses-table
ORDER BY credits DESC
LIMIT 3;
```

courses-table

Course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. Syntax

```
SELECT MAX (grade) AS max-grade,
       MIN (grade) AS min-grade,
       AVG (grade) AS avg-grade
```

FROM enrollment-table;

enrollment-table

max-grade	min-grade	avg-grade
90	78	84.6

10. Syntax SELECT course_id,
 COUNT (enrollment_id) AS enrollment_count
FROM enrollment_table;

enrollment_table

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Syntax SELECT department,
 SUM (salary) AS total_salary,
 SUM (bonus) AS total_bonus
FROM salaries_table;

~~WHERE~~
salaries_table

department	total_salary	total_bonus
IT	122,000	10500
HR	109,000	7500
Finance	70000	6000

12. Syntax SELECT department,
 AVG (salary) AS avg_salary
FROM salaries_table

GROUP BY department;
salaries_table

department	avg_salary
IT	61000
HR	54500
Finance	70000

13. Syntax SELECT employee_id,
name,
salary,
bonus,
SUM(salary + bonus) AS total_compensation
FROM salaries-table
WHERE (salary + bonus) > 60 000;

Salaries-table

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

14. Syntax SELECT department,
SUM(budget) AS total-budget,
AVG(budget) AS avg-budget
FROM projects-table
GROUP BY department;
HAVING avg-budget > 70 000;

projects-table

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	8000

15. Syntax SELECT *
FROM projects-table
WHERE department BETWEEN 50 000 AND 120 000 AND
NOT 'marketing';

project_id	project-name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
5	Dashboard HR Portal	IT HR	50 000